EDS 223 Assignment 2

Aquaculture Analysis

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Background

Marine aquaculture can be a more sustainable option than the alternative land-based meat for the global food supply (Hall et al, 2011). Gentry et al. created a map based on the neccessay constraints for marine aquaculture including, ship traffics, dissolved oxygen, and bottom depth, concluding that the global seafood demand could be met with less than 0.015% of the global ocean area.

Assignment Description

For this assignment, the Exclusive Economic Zone (EEZ) will be used to determine suitable regions throughout the west coast to develop marine aquaculture for several oyster species, as well as a species of choice, which was chosen to be the blue mussel (*Mytilus edulis*).

The main goal is to create a function that will input the following: minimum and maximum sea surface temp, minimum and maximum depth, and the species name. The output will be a map of the EEZ regions colored by the amount of suitable areas for the selected species.

For the oysters, the information is the following:

- **Oysters:**
  - Sea surface temperature: 11-30°C
  - Depth: 0-70 meters below sea level
- **Blue mussel:**
  - Name: *Mytilus edulis*
  - Sea surface temperature: 5-20°C
  - Depth: 0-60 m

Data Preparations

```
# Load libraries
library(tidyverse)
library(sf)
library(here)
library(stars)
library(tmap)
library(janitor)
library(terra)
library(geodata)
library(kableExtra)
```

```
# Read in data
set_file <- c(
    here('data/average_annual_sst_2008.tif'),
    here('data/average_annual_sst_2009.tif'),
    here('data/average_annual_sst_2010.tif'),
    here('data/average_annual_sst_2011.tif'),
    here('data/average_annual_sst_2012.tif')
)

sst_data <- rast(set_file)

depth <- rast('data/depth.tif')

eez_data <- st_read('data/wc_regions_clean.shp')
```

Data Process

```
# Calculate SST mean from 2008-2012 and convert Kelvin to Celsius
sst_mean_c <- mean(sst_data, na.rm = TRUE) - 273.15

# Crop data to match
crop_depth <- crop(depth, sst_mean_c)

# Transfer sst data using bilear method to match
resampled_depth <- resample(crop_depth, sst_mean_c, method = "bilinear")
```

```r
# Verify resolution
  # Checking and converting depth to match SST resolution, extent, and CRS
  if (all(res(sst_mean_c) == res(resampled_depth))) {
    print("Resolutions match!")
  } else {
    warning("Resolutions don't match!")
    res(resampled_depth) <- res(sst_mean_c)
  }

  if (ext(sst_mean_c) == ext(resampled_depth)) {
    print("Extents match!")
  } else {
    warning("Extent don't match!")
    ext(resampled_depth) <- ext(sst_mean_c)
  }
# test if the resampled depth CRS matches the sst_mean_c CRS
  if (crs(resampled_depth) == crs(sst_mean_c)) {
    print("CRS match!")
  } else {
    warning("CRS don't match!")
    crs(resampled_depth) <- crs(sst_mean_c)
  }
```

Workflow for Oysters

```r
# variables for mix and max
min_sst <- 11
max_sst <- 30


# Reclassifying SST and depth data into suitable locations
# SST reclassification matrix
  rcl_sst <- matrix(
    c(
      -Inf, min_sst, 0,
      min_sst, max_sst, 1,
      max_sst, Inf, 0
    ),
    ncol = 3,
    byrow = TRUE
```

```
  )

min_depth <- 0
max_depth <- -70

# Depth reclassification matrix
rcl_depth <- matrix(
    c(
      -Inf, max_depth, 0, # should max_depth be negative?
      max_depth, min_depth, 1,
      min_depth, Inf, 0
    ),
    ncol = 3,
    byrow = TRUE
  )
```

```
# reclassify
zone_sst <- classify(sst_mean_c, rcl = rcl_sst)

zone_depth <- classify(resampled_depth, rcl = rcl_depth)

#combine them the two zones together
depth_sst_zones <- c(zone_sst, zone_depth)
```

```
# Find suitable zones for oysters
suitable_zones <- lapp(depth_sst_zones, function(sst_data, depth_data) sst_data * depth_data
```

```
# Determine suitable areas within EEZ
sf::st_crs(eez_data)<- terra::crs(suitable_zones)
```

```
# Convert to raster
eez_raster <- rasterize(eez_data, suitable_zones, field = "rgn_id")
```

```
# Calculate cell size
cell_size <- cellSize(suitable_zones, unit = "km", mask = TRUE)
```

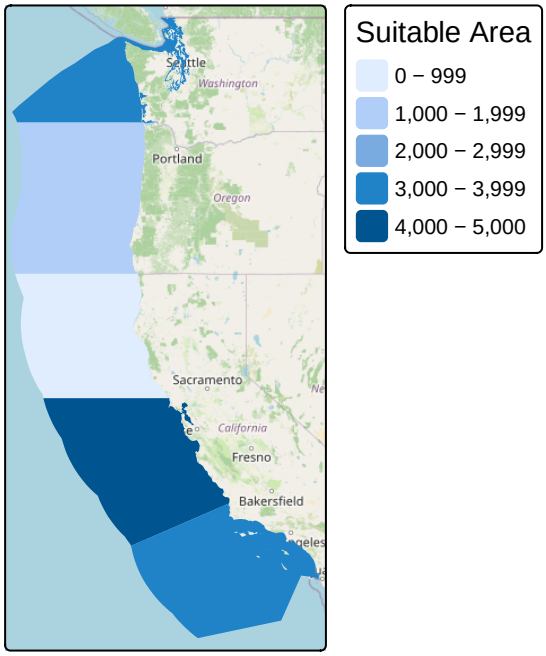
```
# keep cell area only where suitable areas equals 1
suitable_zmask <- mask(cell_size,suitable_zones, maskvalues = 0)
```

```
# use zonal function to sum masked result and eez raster
desired_zone <- zonal(suitable_zmask,eez_raster, fun = 'sum', na.rm = TRUE)
```

```
# use tibble()
z_tibble <- as_tibble(desired_zone)

map_1 <- left_join(eez_data, z_tibble, by = "rgn_id")
```

```
# Visualize EEZ
tm_shape(map_1) +
    tm_basemap("OpenStreetMap") +
    tm_fill(
        fill = "area",
        fill.legend = tm_legend(title = "Suitable Area")
    )
```



Numerical Oyster Reprsentation by Exclusive Economic Zone

```
# Create a table for area by region and suitability
map_1 |>
```

```
  st_drop_geometry() |>
  select(rgn_id, area, area_km2) |>
  arrange(desc(area)) |>
  mutate(pct_suitable = area / area_km2 * 100) |>
  kable(
    digits = 1,
    col.names = c(
      "EEZ Region",
      " Suitable Area (km²)",
      "Total EEZ Area (km²)",
      "Suitable (%)"
    ),
    caption = "Oyster Aquaculture by Exclusive Economic Zone (EEZ) in the West Coast",
    booktabs = TRUE
  ) |>
  kable_styling(latex_options = c("striped", "hold_position"))
```

Table 1: Oyster Aquaculture by Exclusive Economic Zone (EEZ) in the West Coast

EEZ Region	Suitable Area (km²)	Total EEZ Area (km²)	Suitable (%)
3	4373.6	202738.3	2.2
4	3329.9	206860.8	1.6
5	3091.5	66898.3	4.6
1	1395.7	179994.1	0.8
2	323.9	164378.8	0.2

Creat Function to Find Suitable Locations within EEZ West Coast Regions

```
eez_species <- function(
    min_sst,
    max_sst,
    min_depth,
    max_depth,
    species_name
){
# SST matrix reclassification
    rcl_sst <- matrix(
    c(
      -Inf, min_sst, 0,
```

```
      min_sst, max_sst, 1,
      max_sst, Inf, 0
    ),
    ncol = 3,
    byrow = TRUE
    )

# reclassify
    zone_sst <- classify(sst_mean_c, rcl = rcl_sst)

# Depth reclassification matrix
    rcl_depth <- matrix(
    c(
      -Inf, max_depth, 0,
      max_depth, min_depth, 1,
      min_depth, Inf, 0
    ),
    ncol = 3,
    byrow = TRUE
    )

    zone_depth <- classify(resampled_depth, rcl = rcl_depth)

# Combine the two zones together
    depth_sst_zones <- c(zone_sst, zone_depth)

# Find suitable zones
    suitable_zones <- lapp(depth_sst_zones, function(sst_data, depth_data) sst_data * depth_

# Determine suitable zones inside EEZ
    sf::st_crs(eez_data)<- terra::crs(suitable_zones)

# Convert to raster
    eez_raster <- rasterize(eez_data, suitable_zones, field = "rgn_id")

# Calculate cell size
    cell_size <- cellSize(suitable_zones, unit = "km", mask = TRUE)

# Keep cell area only where suitable areas equals 1

    suitable_zmask <- mask(cell_size,suitable_zones, maskvalues = 0)
```

```
# Use zonal function to sum masked result and eez raster

    desired_zone <- zonal(suitable_zmask,eez_raster, fun = 'sum', na.rm = TRUE)

# use tibble()
    z_tibble <- as_tibble(desired_zone)


    map <- left_join(eez_data, z_tibble, by = "rgn_id")

# Visualize EEZ
    tm_shape(map) +
        tm_basemap("OpenStreetMap") +
        tm_fill(
            fill = "area",
            fill.legend = tm_legend(title = "Suitable Area (km^2)")
            ) +
                tm_borders() +
                tmap_options(component.autoscale = FALSE) +
                tm_title(
                 paste(
                "Suitable",
                species_name,
                "Area by Exclusive Economic Zone (EEZ)"
                ))
}
```

EEZ Species Function: Finding Suitable Marine Aquaculture for Given Species

Utilizing the function created above, the map can be easily recreated for the desired blue mussel rather than recreate the lengthy process of manually inputing the inforamtion for sea surface temperature and depth, which leads to the reproduction of matching crs, masking, and more.

The function is called after EEZ species and used as:

- `eez_species(min_sst, max_sst, min_depth, max_depth, species_name)`

Simply Input into the funtion:

- Oysters:
  - min_sst = 11°C
  - min_sst = 30°C

  - min_depth = 0 meters
  - max_depth = -70 meters
  - Species_name = Oysters

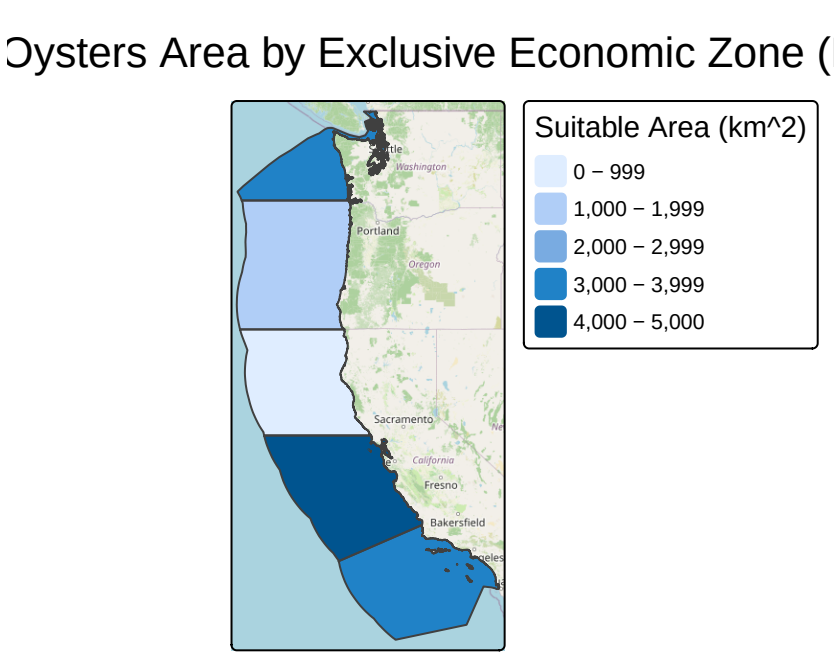
- Blue mussels:

  - min_sst = 5°C
  - min_sst = 20°C
  - min_depth = 0 meters
  - max_depth = -60 meter
  - species_name = Blue Mussel

- Output:

  - map of EEZ regions colored by amount of suitable area

```
# Create a suitable EEZ map for oysters using function
eez_species(11, 30, 0, -70, 'Oysters')
```




```
# Create a suitable EEZ map for mussles using function
eez_species(5, 20, 0, -60, 'Blue Mussel')
```

ue Mussel Area by Exclusive Economic Zone (EEZ)

